

How to Submit New Outlines to GLIMS

NSIDC

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1 Overview

This document is a guide to submitting new glacier outlines to the GLIMS Glacier Database. The GLIMS Glacier Database is a multitemporal geospatial relational database containing glacier outlines and other features related to mapping glacier cover. The database also contains information about who created the outlines, the imagery or maps used to digitize the outlines, and the methods used for mapping, as well as documenting extinct glaciers¹.

¹Extinct glaciers should be a separate submission.

A basic submission consists of one or more glacier outlines for a region, information about the people who mapped the outlines and the imagery used for this mapping.

We describe the required information and file formats in the following sections.

Please submit all files to glacierdata@nsidc.org. The NSIDC team will review the submission and contact you with any questions about your submission or request for further information.

2 Required Information

2.1 Submission Metadata

2.1.1 Metadata Format

Please submit the following metadata as a text file with each metadata element on a separate line.

2.1.2 *Title of Submission*

The submission title helps us track the submission. We suggest a title that includes the name of the submitter, the name of the region being mapped, and the year for which glaciers are mapped.

For example, "Agassiz, glacier outlines of Switzerland, 1840"

2.1.3 *Geographic Region*

The original GLIMS project organized the mapping effort through Regional Centers. GLIMS still uses Regional Center IDs. For your submission please use the Regional Center ID that covers the outlines you are submitting. A list of Regional Center IDs can be found on the [GLIMS Regional Centers](#) page.

We also require the common name of the Geographic Region containing the glacier outlines. For country inventories please use the country name. For multi-national regions, please use the commonly accepted name for this region, for example North America, European Alps, Himalaya, High Mountain Asia. For smaller regions, please give the state, mountain range or other commonly recognized term.

2.1.4 *Personel*

The person making the submission is the *submitter*. Other people who helped map glacier outlines are *analysts*. We would like the name, institution and an email for the *submitter* and any *analyst* that contributed to the work. This allows GLIMS to give credit to the people mapping glaciers and also provides a contact if users of this information have questions about the outlines.

2.1.5 *Files*

Please provide a description of the contents of all files you are submitting. See Accepted File Formats for more information about the format and structure of files.

2.1.6 *Description of Analysis Methods*

Please provide the date on which the analysis was performed. Often analyses take a while to complete. A date to the closest month is OK. For example, if mapping was done in October 2026, enter 2026-10-01.

Describe any image processing that was performed, e.g. image transformations, radiometric corrections, use of band ratios or thresholds.

A brief description of the mapping method is also requested. For example, **Manual digitizing of band ratios of Sentinel-2 images**. If an automated or semi-automated mapping procedure was used please indicate if manual corrections were made and what percentage of outlines were manually corrected.

2.1.7 *Embargo Period*

To encourage timely submission of outlines, glacier outlines submitted to GLIMS can be embargoed for a period of time. The length of this time period is usually 6 months to allow analysts to prepare and submit journal articles.

2.1.8 *Related Publications*

Please provide the citation for any publications related to the submitted outlines. Please provide a DOI if one is available.

2.2 Glacier Outlines and other glacier features

2.2.1 Features in the GLIMS Glacier Database

The majority of features in the GLIMS Glacier Database are glacier outlines. However, the database also contains the outline of debris cover on glaciers; supraglacial and proglacial lakes; center lines of glaciers; and transient snow lines. Glacier outlines are the only **required** features.

2.2.2 Accepted file formats

Glacier features can be submitted as ESRI Shapefiles or GeoJSON. Most submissions use ESRI Shapefile.

2.2.3 Feature Geometry

A glacier outline, debris cover, supraglacial lake or proglacial lake can be either a Polygon or MultiPolygon. Center lines and transient snowlines should be LineStrings or MultiLineStrings

Each type of glacier feature is mapped as a separate vector entity.

Glacier outline features should represent Nunataks² as *holes*. Exposed ice within areas of debris cover and proglacial lakes, and islands in supraglacial lakes may also be represented as *holes*.

Fragmented glaciers can be mapped as MultiPolygons, where each fragment is a polygon in the MultiPolygon collection. Alternatively, fragments can be mapped as separate Polygons. If fragments are mapped as separate Polygon features, all of the fragments of the same glacier must have the same id (see feature attributes)³.

Glacier outlines, debris cover, supraglacial lakes and proglacial lake features may be combined in a single shapefile. If multiple feature types are included in the same shapefile, each feature must have a category attribute. See Section [2.2.4](#).

²A Nunatak is a mountain, or any exposed ground or bedrock, that projects from and is surrounded by glacier ice.

³A glacier id can be an arbitrary identifier assigned by an analyst or an identifier assigned as part of a local glacier inventory. GLIMS assigns GLIMS Glacier IDs to glaciers. However, the GLIMS glacier database also stores local identifiers.

2.2.4 Attributes

Table 1 lists the attributes that can be assigned to individual glacier features (outlines, debris cover, etc). Required attributes are indicated in the “Required” column. If outlines are mapped from satellite or aerial imagery, attributes marked “Y*” are required. If outlines are mapped from maps, attributes marked “Y^” are required.

Table 1: Attributes for glacier features

Attribute Name	Required	Description	Data Type	Example
name	N	Glacier Name	string	"Unteraargletscher"
local_id	Y	ID assigned to glacier by analyst or by regional inventory	string	"0001"
wgms_id	N	ID assigned by World Glacier Monitoring Service	string	
inst_name	Y*	Recognized name of instrument.	string	"Sentinel-2", "SuperDove"
orig_id	Y*	ID for image assigned by image provider	string	"S2B_MSIL2A_20250722T102559_N0511_R108_T3"
acq_time	Y*	Date of image acquisition in ISO 8640 format	string	2026-08-13 or 2026-08-13 13:15:00
map_name	Y^	title or name of map	string	"Zermatt Gornergrat"
auth_pub	Y^	Author or publisher of map	string	"Bundesamt fur Landestopographie"
pub_date	Y^	Date map published	string	"1998"
scale	Y^	Scale of map	string	"1:25000"
analyst_name	Y	Name of analyst who created outline	string	"Agassiz, Louis"
category	N	Category of feature	string	"glac_bound"

If the outlines shapefile contains named and unnamed glaciers, the value of `name` for unnamed glaciers should be an empty string, e.g. “”.

Table 2 list the valid values for the `category` attribute. The table also lists the accepted geometry types for each feature. A `category` attribute is only required if multiple glacier features are submitted.

Table 2: Valid values for `category` attribute

category value	Description	Feature Types
“glac_bound”	Glacier boundary	Polygon or MultiPolygon with holes
“centerline”	Glacier centerline	LineString or MultiLineString
“snow_line”	Location of snow line	LineString or MultiLineString
“pro_lake”	Proglacial Lakes	Polygon or MultiPolygon with holes
“supra_lake”	Supraglacial Lake outline	Polygon or MultiPolygon with holes
“debris_cov”	Debris covered glacier ice	Polygon or MultiPolygon with holes